

Haihao Lin

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Research interests: VLA post-training and representation stability; real-robot learning; cross-task generalization; deployment-aware model adaptation.

Education

University of Chinese Academy of Sciences (UCAS) Sep. 2024–Jun. 2027 (expected)
M.S. in Computer Science (Artificial Intelligence). Advisor: Prof. Xiangsheng Huang.

Jiangsu University Sep. 2020–Jun. 2024
B.E. in Computer Science and Technology. GPA rank: **1/171**.

Research

FiberTune: Preserving Action-Fiber Visual Residuals in VLA Fine-Tuning

First and corresponding author | Manuscript under review | [arXiv:2606.08653](#) | [Project: fibertune.github.io](#)

- FiberTune addresses visual representation degradation and cross-task generalization loss during action-supervised VLA fine-tuning. It uses an online action probe to identify action-predictive directions, projects them out of intermediate visual representations for the auxiliary objective, and applies frozen-teacher alignment and spectral-entropy regularization to the resulting residual, adding no inference-time overhead.
- Validated with $\pi 0.5$ and OpenVLA-OFT on CALVIN and LIBERO, plus a real-world SO-101 task. FiberTune outperformed task-loss-only fine-tuning in all six controlled settings; CALVIN ABC→D SR(5) increased from 61.4% to 72.1%, and SO-101 task success from 72.7% to 78.1%.
- Built the SO-101 data-collection and OOD evaluation pipeline and ran controlled training, evaluation, and representation diagnostics for $\pi 0.5$ and OpenVLA-OFT across CALVIN, LIBERO, and real-robot settings.

Current Research

- Experiments on representation and control interfaces across VLAs and WAMs, using FastWAM and related models to study cross-task generalization and real-robot learning.

Robotic Systems

ROS2 Systems Engineering for a Commercial Cleaning Robot Mar. 2026–Jul. 2026

- Designed and implemented autonomous docking, charging alignment, fall prevention, and a companion docking simulation system, covering sensor selection and placement, charger communication, state monitoring, and docking control; performed simulation-based parameter tuning and stress testing, ROS2 deployment on the real robot, and repeated closed-loop trials.
- Developed the initial coverage-path implementation, adapted an external algorithm, and integrated it on the real robot; optimized the wake-word and speech-recognition pipeline for noisy onboard conditions and contributed to system-level state-machine integration.

Intelligent Logistics Robot System Design and Prototype Validation Aug. 2025–Jan. 2026

- Built an xArm + RGB-D tabletop manipulation prototype and deployed it on Jetson, covering hand-eye calibration, suction/gripper integration, logistics use-case analysis, and Web API prototyping.

Additional Research and Engineering

Confidential Multi-Objective Task-Planning and Resource-Allocation System Mar. 2024–Oct. 2024

- Implemented multiple planning and scheduling algorithms, randomized data generators for multiple scenarios, and the full test framework; consolidated the team’s technical proposal and bid materials and delivered the on-site presentation.

AI-Enabled Pharmaceutical Property Prediction and Screening Sep. 2023–Jun. 2024

- Completed data processing, model reproduction, property prediction, and the screening pipeline for the large-molecule workstream of an AI-enabled pharmaceutical platform, while contributing to API integration; an undergraduate thesis based on this work received a university outstanding-thesis award.

MaaAssistantArknights Open-Source Automation Project Jun. 2022–Apr. 2026 | GitHub 20k+ stars

- Early key contributor; built multiple early features across the C++ core and WPF GUI while implementing and continuously optimizing a complex planner for mutual-exclusion constraints.

Skills

Programming and frameworks: Python, PyTorch, LeRobot, and RLinf.

Embodied models: the π family ($\pi 0.5$), the OpenVLA family (OpenVLA-OFT), SmolVLA, and FastWAM.

Robotics and systems: ROS2, C++/CMake, OpenCV, RealSense/RGB-D, xArm SDK, Jetson/RK, and Linux/Docker.

Honors and Competitions

- National Scholarship (twice); Jiangsu Provincial Merit Student; Jiangsu University President’s Medal; Outstanding Graduate; Outstanding Undergraduate Thesis.
- Three silver medals at ICPC Asia Regional Contests; highest-ranked bronze-medal team at the 2024 ICPC Asia-East Continent Final; two national first prizes in the Lanqiao Cup (C/C++ and Python), including 1st place in Jiangsu and 9th nationally in Python.